Today, I discuss *one* of the many ways to implement a GraphRAG. My current approach uses AWS Neptune Analytics - “a *memory-optimized* graph database engine for analytics”[1]
Additionally - and crucially, it can *also* serve as a vector data store, making it a single-shot solution for storing a GraphRAG and a VectorRAG[2].
The graph data loaded into Neptune Analytics *can* be queried/subject to Exploratory Data Analysis using openCypher queries[3] running in a Jupyter Notebook. This is made possible by setting up the Jupyter Notebook in AWS Sagemaker to interact with the graph data. AWS Neptune Analytics also leverages an open-source graph-explorer[4] that allows for visualization of the underlying graph that powers the GraphRAG.
While it took a couple of iterations to get the workflow right, here is what finally worked for me:

1. Setting up the networking aspect correctly:
I created a VPC with 4 subnets (2 public and 2 private). I also had a NAT gateway and an Internet Gateway available.
I created the Neptune Analytics instance within this VPC (inside one of the private subnets) and then created a Jupyter notebook (accessible via AWS Sagemaker) *within the same VPC* as the Neptune Analytics instance. (I actually created it within the same private subnet).
Next, I created security groups (one for AWS Sagemaker and one for the Neptune Analytics instance) that would allow them to communicate. I also created a security group for my laptop’s IP to allow the browser running the graph-explorer instance to retrieve data from the Neptune Analytics instance over port 443.

2.Setting up the KnowledgeBase in AWS Bedrock to use the *existing* Neptune Analytics instance:
I uploaded 2 documents [Amazon-2024-Annual-Report and Amazon-2025-Proxy-Statement.pdf] to an S3 bucket and used this bucket as the data source for the Knowledge Base. Titan Embeddings V2 was the model chosen to generate the vector (and graph) embeddings and AWS Nova Pro was chosen to cover the human interaction piece in this RAG approach.
I found that adding the documents one-by-one seemed to make the ingestion process go without hiccups.
By default, AWS Bedrock creates the Neptune Analytics instance in the default VPC and since my networking piece was set up differently, I set it up to instead use the one that I had just created.

3. Validating the existence of the graph, using the Jupyter Notebook:
The Neptune graph-notebook magic %%oc is a great asset to run openCypher queries on the underlying graph. Below are screenshots of some of the openCypher queries:

```
%%oc
MATCH (n)
RETURN labels(n), count(n)
```


```
{
  "results": [
    {
      "labels(n)": [
        "Entity"
      ],
      "count(n)": 1319
    },
    {
      "labels(n)": [
        "Chunk"
      ],
      "count(n)": 641
    },
    {
      "labels(n)": [
        "DocumentId"
      ],
      "count(n)": 2
    }
  ]
}
```


```
%%oc
MATCH (c:Chunk)-[r]->(e:Entity)
RETURN
    c.chunk_text AS SourceText,
    type(r) AS Relationship,
    id(e) AS EntityID,
    e.name AS EntityName,
    e.type AS EntityType,
    e.text AS AltText
LIMIT 10
```

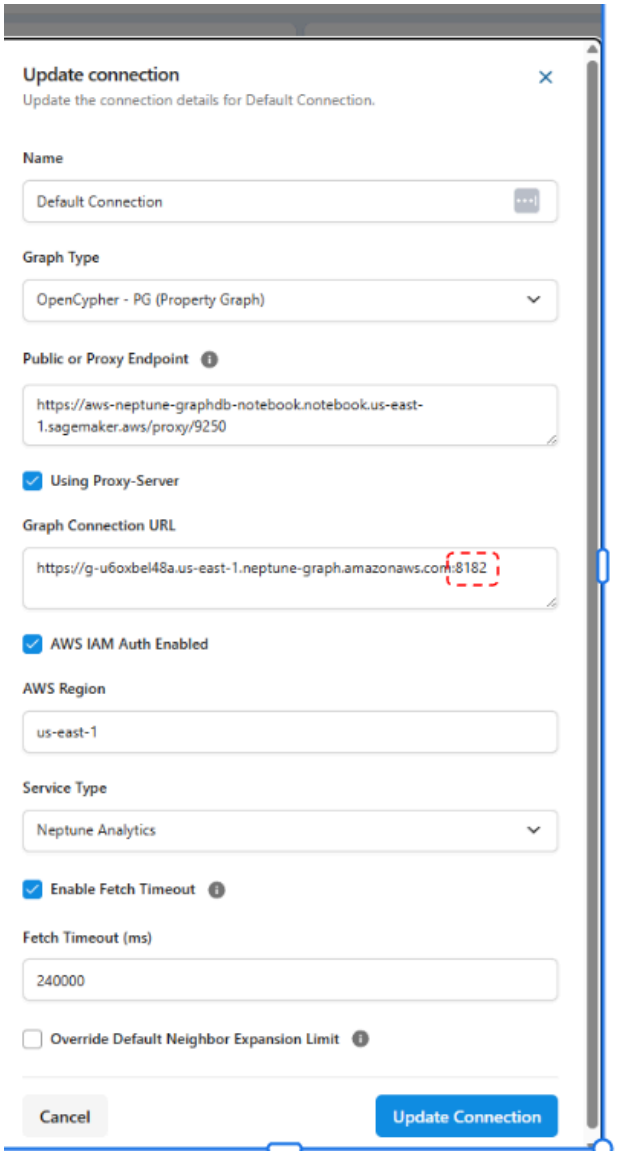
SourceText	Relationship	EntityID
We strive to be Earthâ€™s best employer. We rely on numerous and evolving initiatives to implement this objective and invent mechanisms for talent development, including competitive pay and benefits, flexible work arrangements, and skills training and educational programs such as Amazon Career Choice (education funding for eligible employees). Over 240,000 Amazon employees around the world have participated in Career Choice. We also continue to inspect and refine the mechanisms we use to hire, develop, evaluate, and retain our employees. In addition, safety is integral to everything we do at Amazon and we continue to invest in safety improvements such as capital improvements, new safety technology, vehicle safety controls, and engineering ergonomic solutions. Our safety team is dedicated to using the science of safety to solve complex problems and establish new industry best practices. We also provide mentorship and support resources to our employees, and have deployed numerous programs that advance employee engagement, communication, and feedback. 4Available Information Our investor relations website is amazon.com/ir and we encourage investors to use it as a way of easily finding information about us. We promptly make available on this website, free of charge, the reports that we file or furnish with the Securities and Exchange Commission (â€œSECâ€), corporate governance information (including our Code of Business Conduct and Ethics), and select press releases.	CONTAINS	x-amz-bedrock-kb-amazon career choice
Cash provided by (used in) investing activities corresponds with cash capital expenditures, including leasehold improvements, incentives received from property and equipment vendors, proceeds from asset sales, cash outlays for 22acquisitions, investments in other companies and intellectual property rights, and purchases, sales, and maturities of marketable securities. Cash provided by (used in) investing activities was $(49.8) billion and $(94.3) billion in 2023 and 2024, with the variability caused primarily by purchases, sales, and maturities of marketable securities and cash capital expenditures. Cash capital expenditures were $48.1 billion, and $77.7 billion in 2023 and 2024, which primarily reflect investments in technology infrastructure (the majority of which is to support AWS business growth) and in additional capacity to support our fulfillment network. We expect cash capital expenditures to increase in 2025, primarily driven by investments in technology infrastructure. We made cash payments, net of acquired cash, related to acquisition and other investment activity of $5.8 billion and $7.1 billion in 2023 and 2024. We funded the acquisition of 1Life Healthcare, Inc. (One Medical) in 2023 with cash on hand. In Q3 2023, we invested $1.25 billion in a convertible note from Anthropic, PBC.	CONTAINS	x-amz-bedrock-kb-one medical

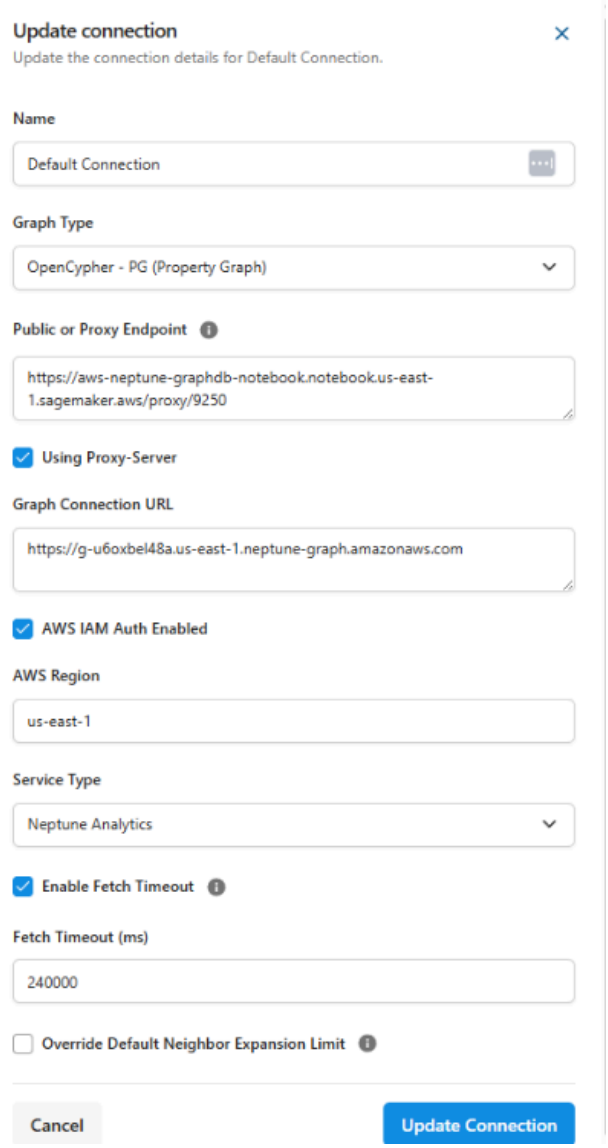
```
%%oc
MATCH (c:Chunk)-[r]->(e:Entity)
RETURN
    c.chunk_text AS SourceText,
    type(r) AS Relationship,
    replace(id(e), 'x-amz-bedrock-kb-', '') AS EntityName,
    labels(e) AS Category
LIMIT 10
```

SourceText	Relationship	EntityName	Category
We strive to be Earthâ€™s best employer. We rely on numerous and evolving initiatives to implement this objective and invent mechanisms for talent development, including competitive pay and benefits, flexible work arrangements, and skills training and educational programs such as Amazon Career Choice (education funding for eligible employees). Over 240,000 Amazon employees around the world have participated in Career Choice. We also continue to inspect and refine the mechanisms we use to hire, develop, evaluate, and retain our employees. In addition, safety is integral to everything we do at Amazon and we continue to invest in safety improvements such as capital improvements, new safety technology, vehicle safety controls, and engineering ergonomic solutions. Our safety team is dedicated to using the science of safety to solve complex problems and establish new industry best practices. We also provide mentorship and support resources to our employees, and have deployed numerous programs that advance employee engagement, communication, and feedback. 4Available Information Our investor relations website is amazon.com/ir and we encourage investors to use it as a way of easily finding information about us. We promptly make available on this website, free of charge, the reports that we file or furnish with the Securities and Exchange Commission (â€œSECâ€), corporate governance information (including our Code of Business Conduct and Ethics), and select press releases.	CONTAINS	amazon career choice	['Entity']
Cash provided by (used in) investing activities corresponds with cash capital expenditures, including leasehold improvements, incentives received from property and equipment vendors, proceeds from asset sales, cash outlays for 22acquisitions, investments in other companies and intellectual property rights, and purchases, sales, and maturities of marketable securities. Cash provided by (used in) investing activities was $(49.8) billion and $(94.3) billion in 2023 and 2024, with the variability caused primarily by purchases, sales, and maturities of marketable securities and cash capital expenditures. Cash capital expenditures were $48.1 billion, and $77.7 billion in 2023 and 2024, which primarily reflect investments in technology infrastructure (the majority of which is to support AWS business growth) and in additional capacity to support our fulfillment network. We expect cash capital expenditures to increase in 2025, primarily driven by investments in technology infrastructure. We made cash payments, net of acquired cash, related to acquisition and other investment activity of $5.8 billion and $7.1 billion in 2023 and 2024. We funded the acquisition of 1Life Healthcare, Inc. (One Medical) in 2023 with cash on hand. In Q3 2023, we invested $1.25 billion in a convertible note from Anthropic, PBC.	CONTAINS	one medical	['Entity']

Note: The results of the openCypher queries (RHS) were exported as csv files before taking the screenshots

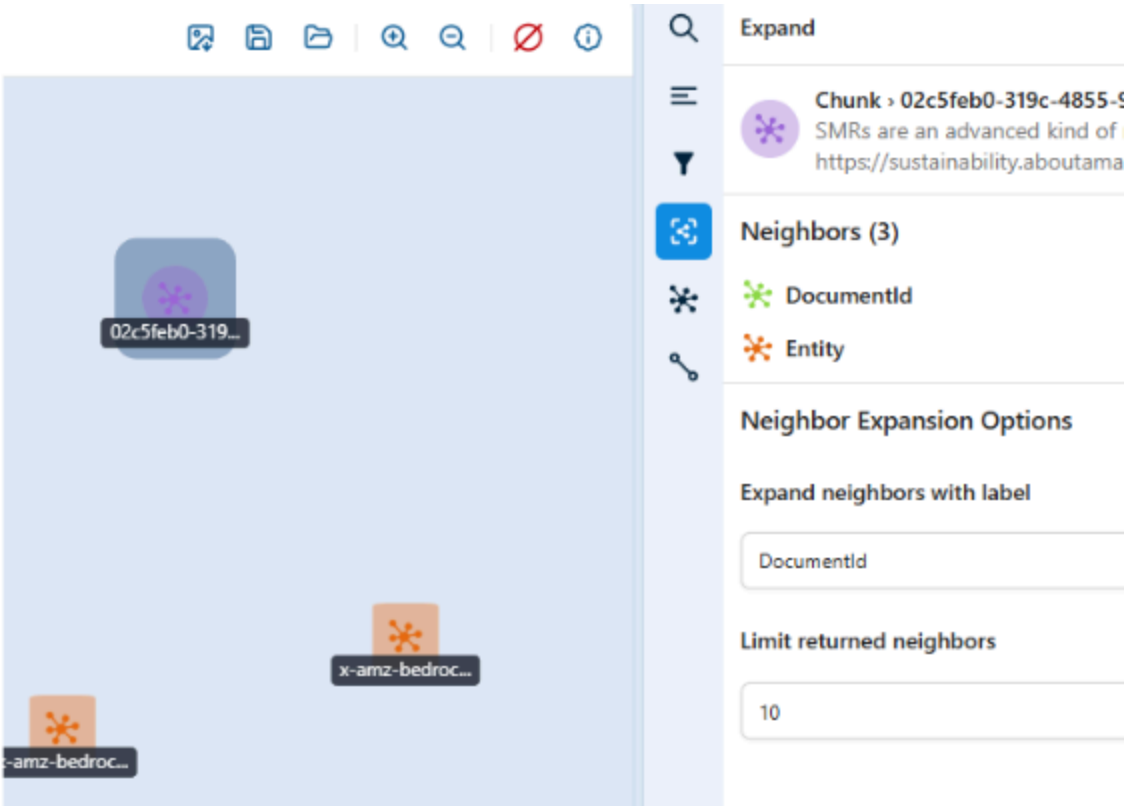
4. Manually updating the proxy URL for the graph explorer:
By default the graph explorer has port 8182 appended to the proxy URL as shown below.



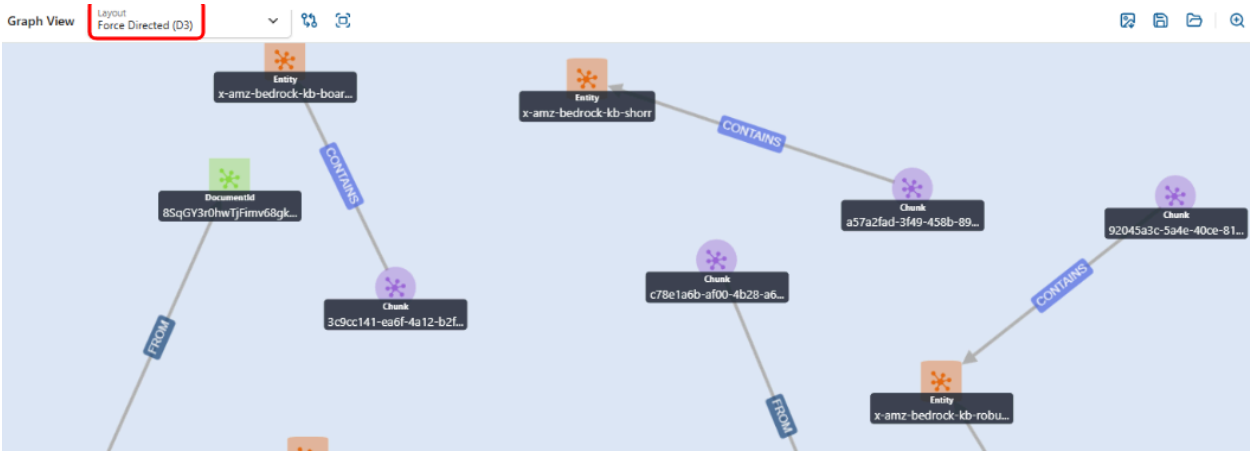


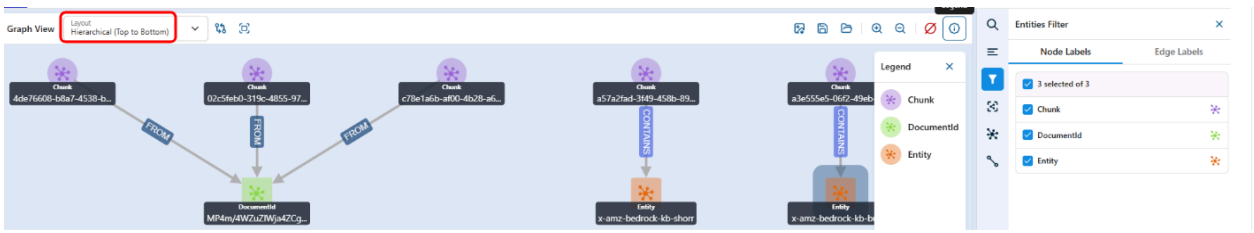
But since it is interacting with a Neptune Analytics instance, that port number does not apply and getting rid of it allowed the graph to be displayed.
Note: Cloudwatch logs are invaluable in figuring out the underlying issues. The graph-explorer merely provides a spinner with a message that just says 'Synchronizing' and it is hard to gauge how long this process needs to run till a graph shows up.

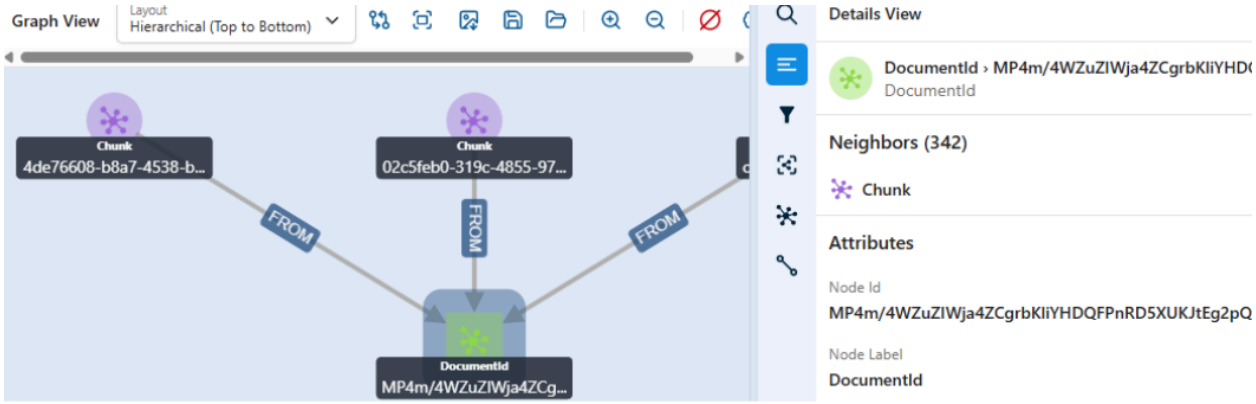
When the graph does come up, it initially shows up *without* the relationship connectors. And all the nodes (document, chunk, entity) are rendered in the same color. However, the UI of the graph-explorer is pretty intuitive and the above can quickly be remedied:



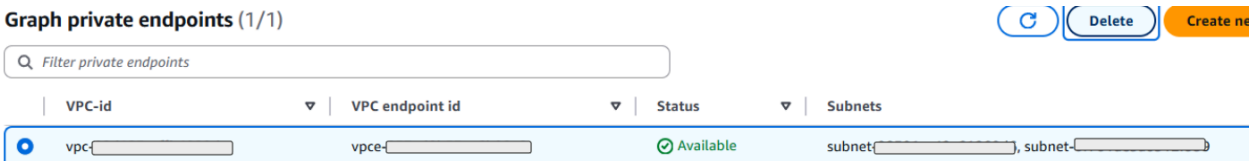
As shown below, it is also possible to tweak the layout of the graph







5. Cleanup is (ALWAYS) super-important:
The Neptune Analytics instance starts accruing charges pretty quickly, but the underlying graph cannot be deleted till the private endpoints that it supports are first deleted.



My Takeaways:
AWS Bedrock and AWS Neptune Analytics make it pretty simple to not only create a GraphRAG but also to visualize and interact with it. However, the underlying networking piece needs to be set up correctly first before *any* of this can proceed successfully.

References

[Neptune Analytics]

[Build GraphRAG applications using Amazon Bedrock Knowledge Bases | Artificial Intelligence]

[Accessing the Neptune Graph with openCypher]

[The open-source graph-explorer - Amazon Neptune]